Navigation in unknown off-road environment using Interval Analysis

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Context

Outdoor unknown off-road navigation stakes in mobile robotics applications are multiples :

- The need to specify **ground geometry**, its nature (e.g., slipping ground), but also the presence and nature of **dynamic objects** to adjust the cartography models and potentially to avoid it when path generation is produced.
- The **change in the environment** (change in luminosity, rain, fog, etc) does not always allow the use of the same sensors continusously. A choice in the sensors but also multisensor fusion, is consequently essential.
- The **noisy nature of acquisition sensors** must be a the subject of careful consideration to bring robust information (possibly guarenteed using Interval Analysis) on which path generation can be based. The context of the described environment corresponds to applications such as construction or farming sites when vehicles evolve jointly with human operators, applications when the security is the top priority.

A robust special sensor : The micro-wave RADAR

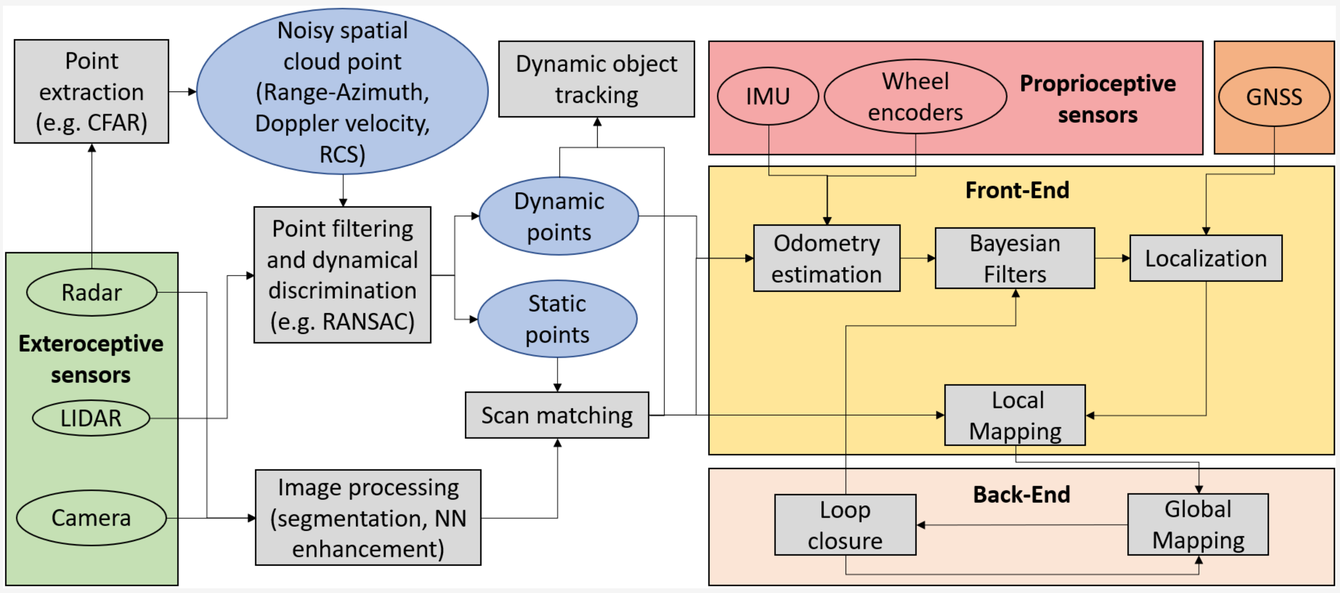

Figure 1. Overview of the radar integration in SLAM.

Similar to the LiDAR, the RADAR is an electromagnetic-based active sensor, which returns the distance to an obstacle with the time of flight between signal transmission and reception. The difference is that the wave frequencies are in the Radio frequency area, between 3 MHz and 300 GHz (76 - 81 GHz for automotive RADAR), whereas LiDAR wave frequency is in the infrared area, approximately 10,000 times higher. This difference in wave frequency translates into the sensor's property where RADAR :

- Is less sensitive to signal attenuation in the atmosphere, allowing **higher maximum range and robustness** to weather conditions such as rain or fog.
- Return the **Doppler velocity** of obstacles.
- Can possibly **go through matter** depending on the type of RADAR
- Is **cheaper** than LiDAR, so attractive for industrial applications

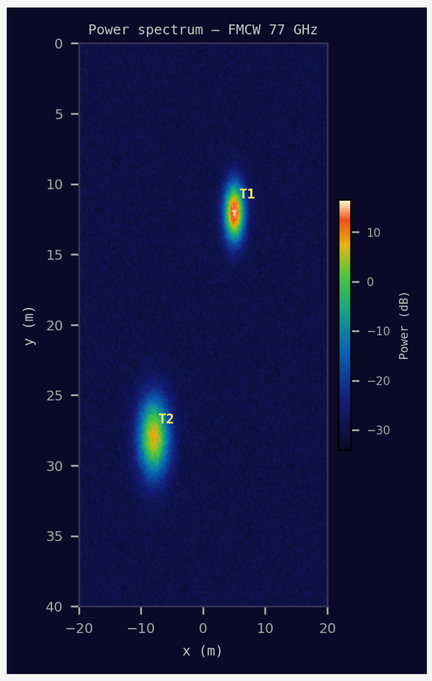

Figure 2. Exemple of RADAR output.

However, the RaDAR drawback also include disadvantages compared to the radar :

- The RADAR has a **poor accuracy** of detections compared to the LiDAR necessitating deeper noise consideration.
- It has a **weak spatial resolution**, which makes it complicated to detect small detection for weak Radar Cross Section target (RCS) depending on shape, geometry and material.

So if the RADAR appears to be a cheap and robust sensors for hostile environment conditions it is complicated to use it without signal processing and/or without joining other sensors [1].

An application of the radar benefit : The guarenteed dynamic tracking

Dynamic objects in environment need to be identified to

- Remap correctly the map that has changed.
- Avoid dynamic obstacles.

The dynamic obstacles are of two types :

- Semi-static where the object does not change its position often (e.g., a door).
- Highly dynamic where the object often move (e.g., a person).

Classically, when range-azimuth sensors are used, dynamic object tracking is performed by comparison of multiple frame to be able to detect semi-static and high dynamicity. When Doppler velocity is found, instantaneous velocity, including angular speed, can be found with the Instantaneous Center of Rotation (ICR) [2]. This kinematics parameters can be found using Interval Analysis taking into accound bounded RADAR noise.

$$y_r \cos(\theta_i) - x_r \sin(\theta_i) = y_{ICR} \cos(\theta_i) - x_{ICR} \sin(\theta_i) - v_{d_i} \cdot \omega^{-1} \quad (1)$$

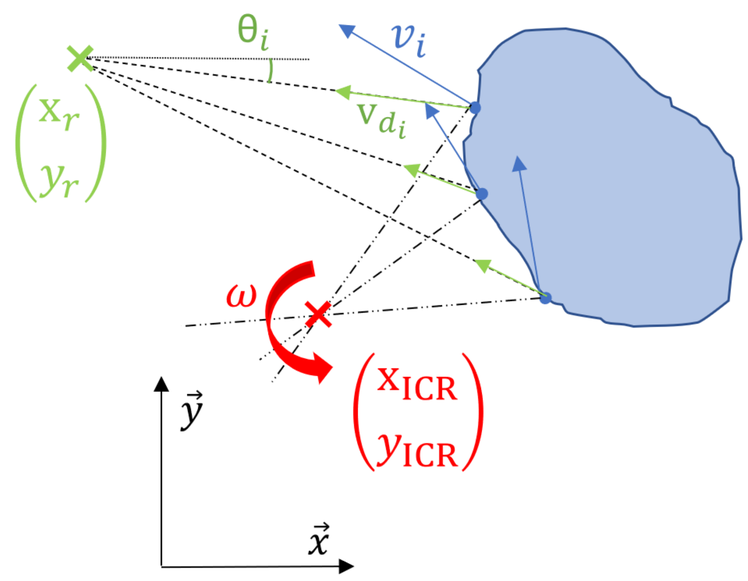

Figure 3. ICR and angular velocity retrieval with distribution of Doppler radial velocity.

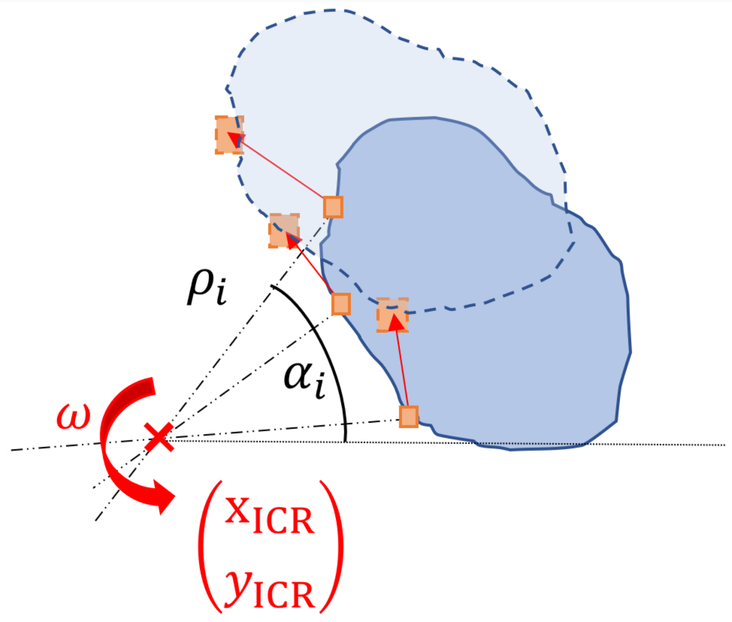

Figure 4. Current measured point projection with ICR and angular velocity estimation.

Research tracks

- Shape estimation over time using Doppler Velocity in addition of spatial information.
- Conditions on the observation of dynamic object (e.g. Visibility, non traversability of objects).

Bibliography

[1] Kyle Harlow et al. "A New Wave in Robotics : Survey on Recent MmWave Radar Applications in Robotics". In : *IEEE TRANSACTIONS ON ROBOTICS* 40 (2024), p. 4544-4560.

[2] Dominik Kellner, Michael Barjenbruch et Jens Klappstein. "Tracking of extended objects with high-resolution Doppler radar". In : *IEEE Transactions on Intelligent Transportation Systems* 17 (2015), p. 1341-1353.